



GOA UNIVERSITY

F.E (NEP) (Sem-I-Civil/COMP/CSE(AI/ML)/IT/ITE/VLSI) & (Sem-II-Mechanical/Mechanical & Automation/ E & E/ ETC/ECOM/ECS) (Revised Course 2024-25)

Paper Title: **Creative Thinking and Innovation**

Paper Code: **AEC-151**

Duration: **2:00** Hours

Total Marks: **30**

Instructions:

- i) Answer any two questions from Part A
 - ii) Answer any two questions from Part B
 - iii) Answer any one question from Part C
- Assume relevant data wherever necessary

Part A

- Q1 a) How do creative thinking approaches contribute to developing innovative products and processes? [2]
- b) Use brainstorming techniques to propose ideas for improving waste segregation and recycling in your community. [2]
- c) How do societal, technological, and business factors shape and influence the process of creativity? [2]
- Q2 a) A construction firm is preparing to begin a large housing project. Create a checklist to help engineers ensure all environmental and safety norms are met before project initiation. [2]
- b) Analyse the role of technology push and market pull in helping a company achieve and sustain competitive advantage. [2]
- c) Why is accurate problem definition important in the problem-solving process? [2]
- Q3 a) A community is facing the issue of frequent water wastage caused by burst water pipes. Apply the steps of the Creative Engineering Design Process to develop an innovative and sustainable solution. [3]
- b) A coastal town is facing issues with waste management. Apply Kipling's Method to develop an effective waste disposal plan. [3]

